Ryerson University

Department of Electrical, Computer, & Biomedical Engineering
Faculty of Engineering & Architectural Science

Course Title	**ELE827**
Course Name	**CMOS Mixed Signal**
Semester/Year	**F2026**

Instructor	**Dr Fei Yuan**

Lab Number	**Lab 1**
Lab Name	**Clock Generator**

Submission Date	
Due Date	

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List of Abbreviations

ADE — Analog Design Environment

ADE XL — Analog Design Environment Explorer

TB — Testbench

TRAN — Transient Analysis

DC — Direct Current (DC analysis / operating point)

PVT — Process / Voltage / Temperature

TT / FF / SS — Typical / Fast / Slow process corners

VDD — Supply Voltage

GND — Ground

VIN — Input Voltage

VOUT — Output Voltage

Schematics and Symbols

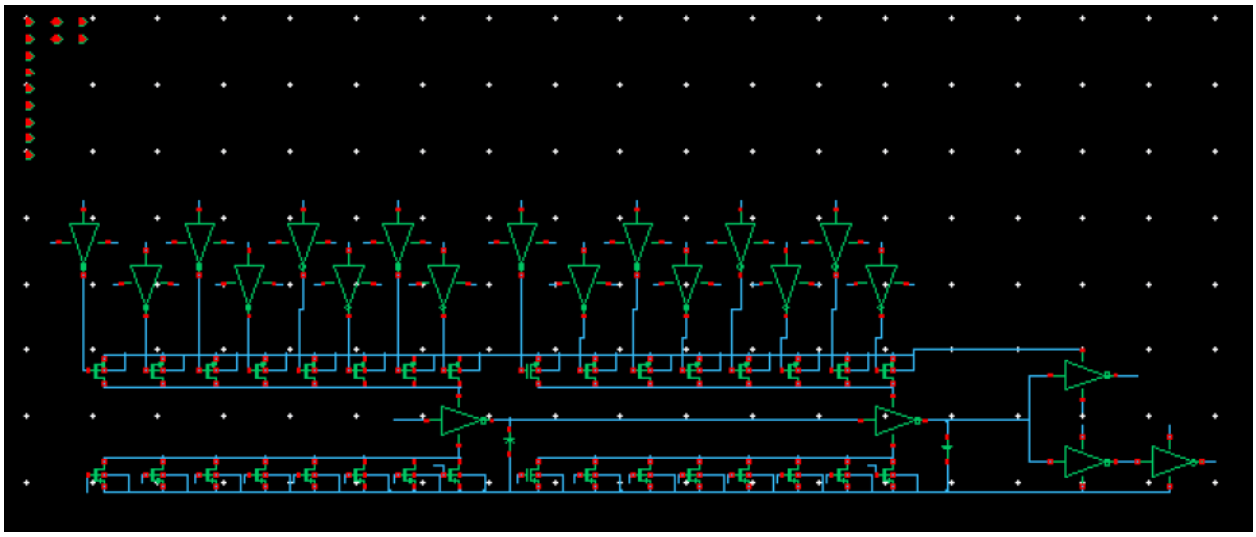


Figure 1: Singled Tuned Delay Stage Schematic

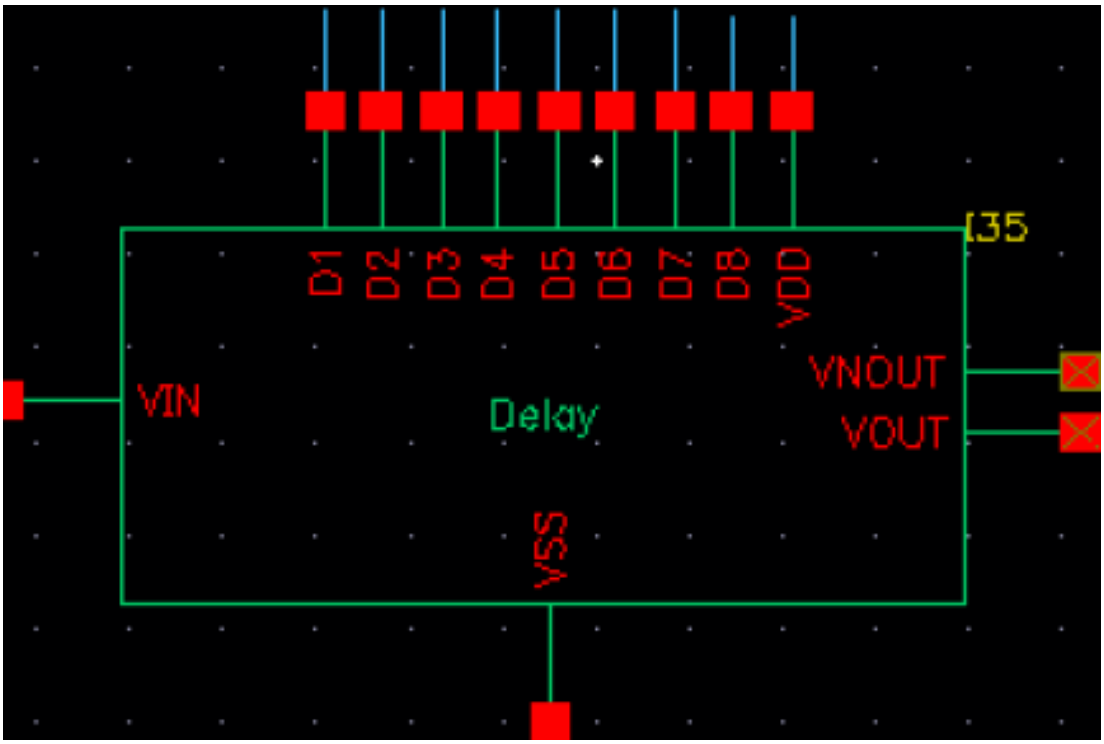


Figure 2: Singled Tuned Delay Stage Symbol

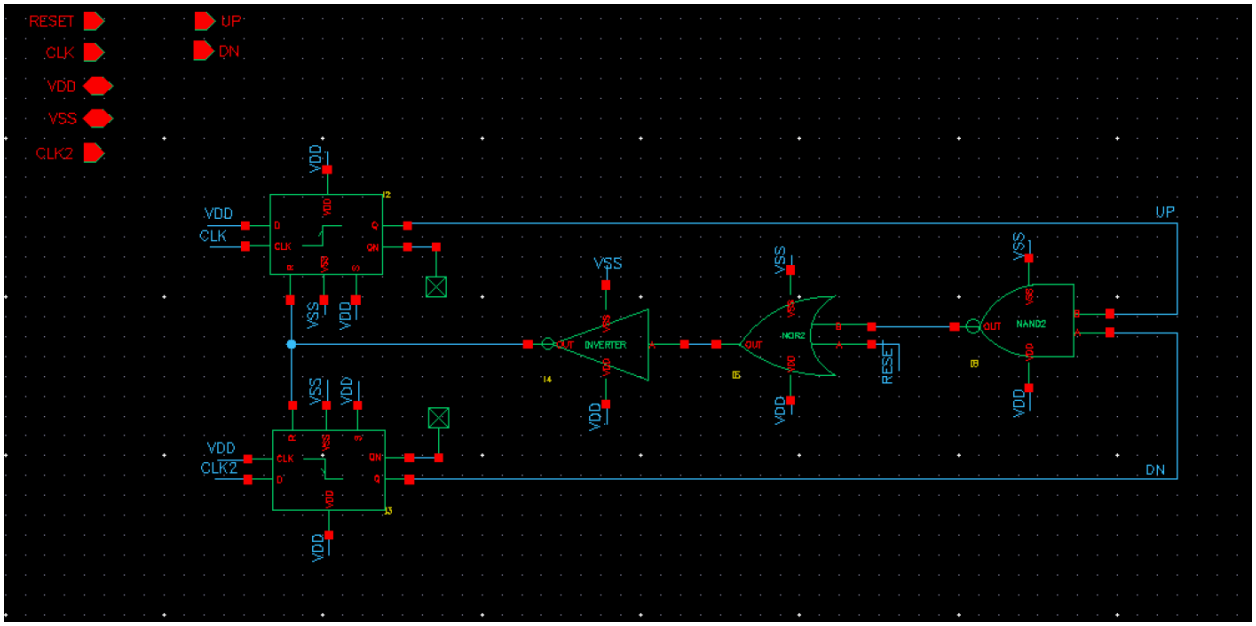


Figure 3: Phase Detector Schematic

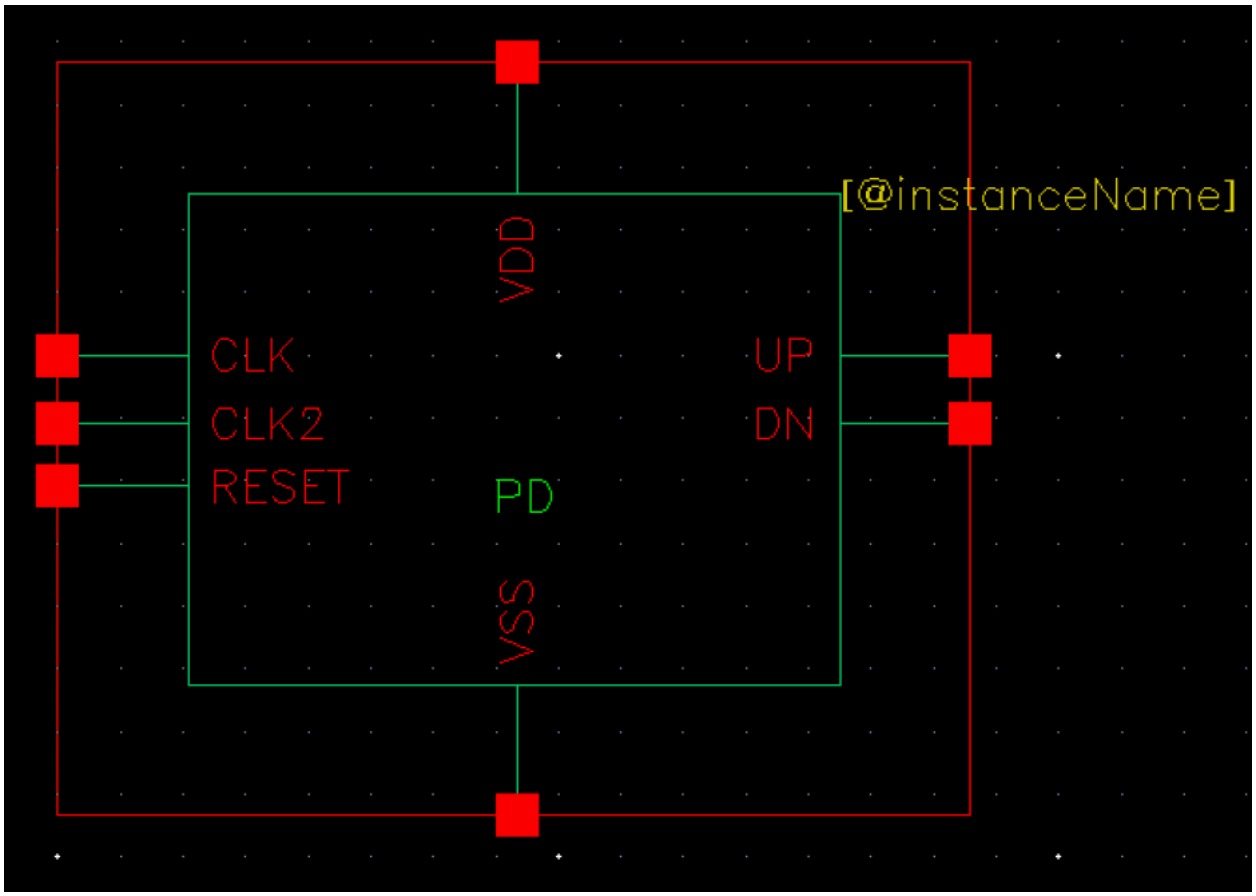


Figure 4: Phase Detector Symbol

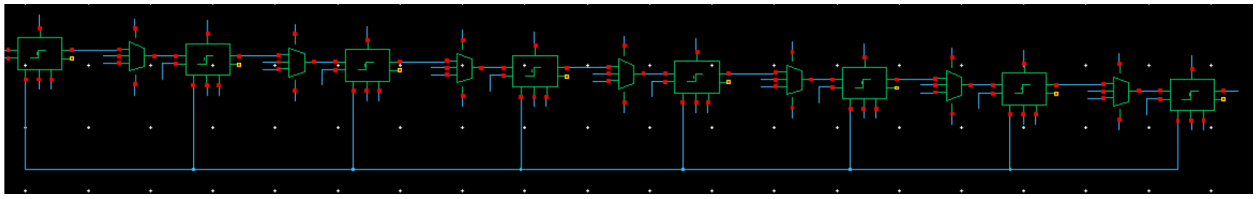


Figure 5: Thermo UP/DN Counter Schematic

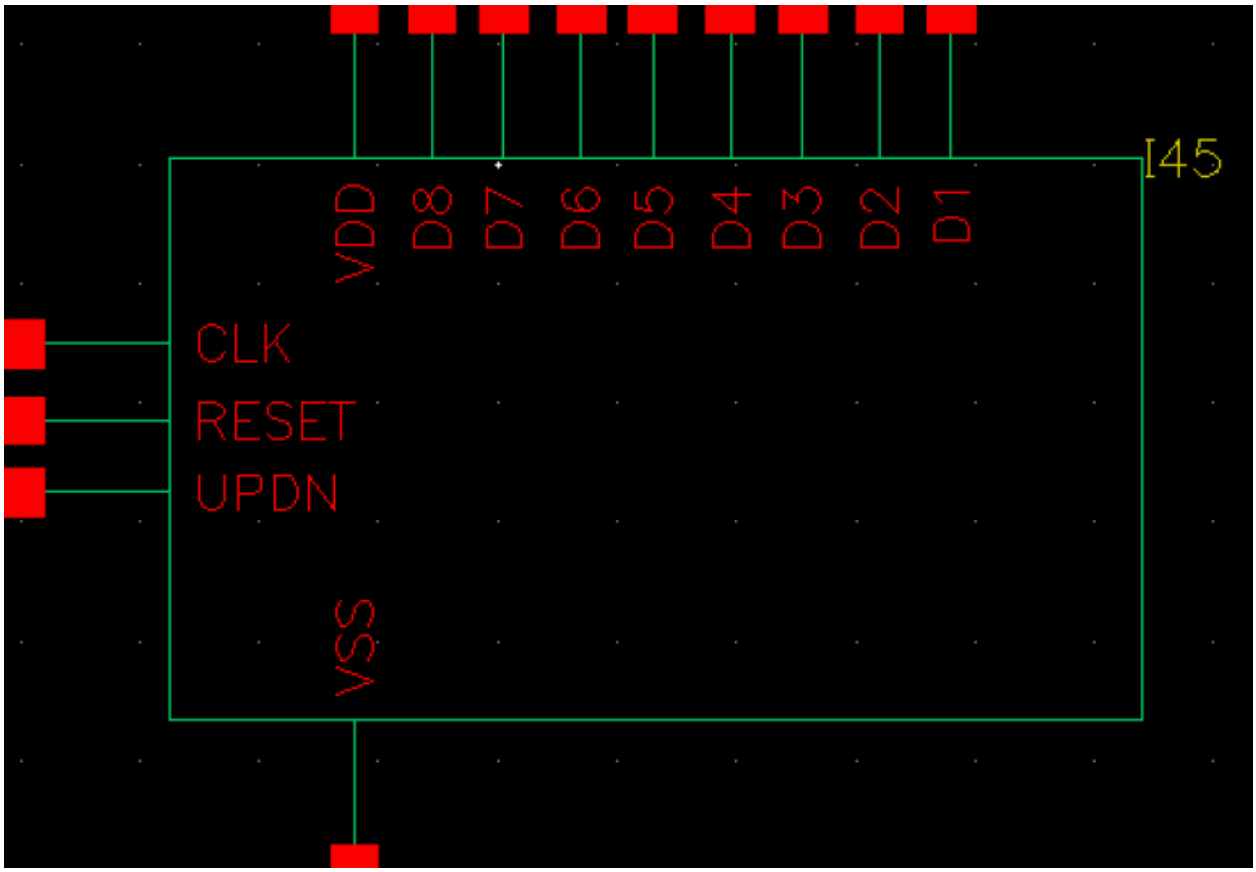


Figure 6: Thermo UP/DN Counter Symbol

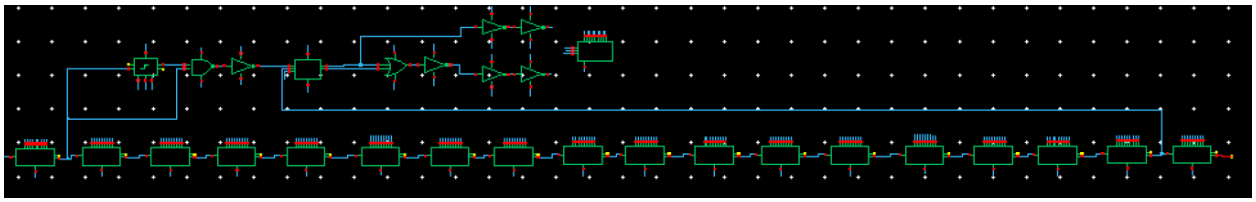


Figure 7 :DLL Schematic

DELAY SIMULATIONS (Incrementing Left Bit 1 by 1 from 0000 0000 with input at the very bottom)

Mapping Schematic (Ideal driver was used to sweep through the bits)

Table 1: Mapping Schematic

Mapping Schematic (Code to Tuning Bits)	
0	0000 0000
128	1000 0000
192	1100 0000
224	1110 0000
240	1111 0000
248	1111 1000
252	1111 1100
254	1111 1110
255	1111 1111

27 C TT

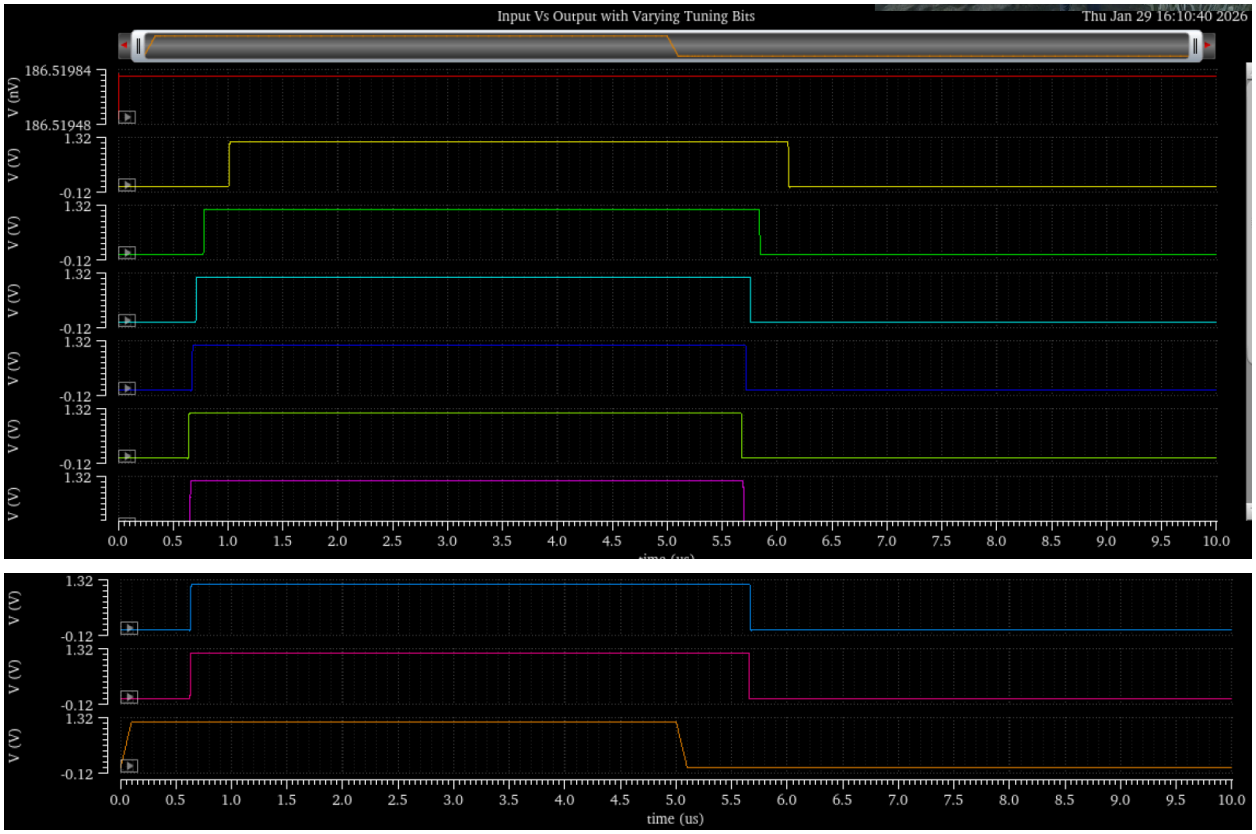


Figure 8: Output of Delay Stages vs Input with varying tuning bits

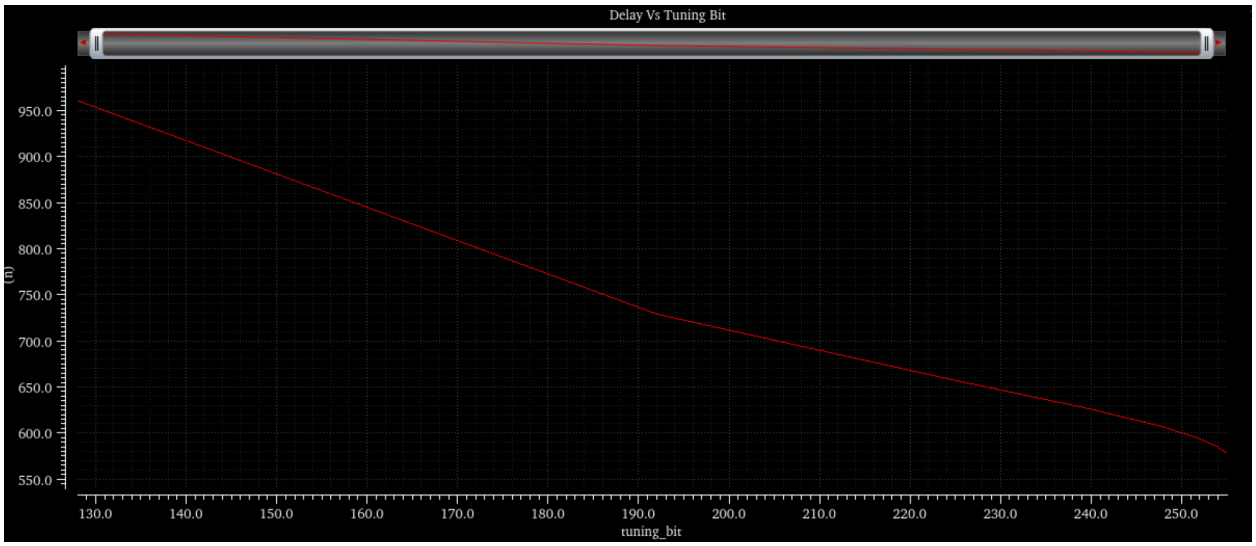


Figure 9: Delay vs Tuning Code

FF -20C

Typo(Input Vs Output with varying tuning bits)

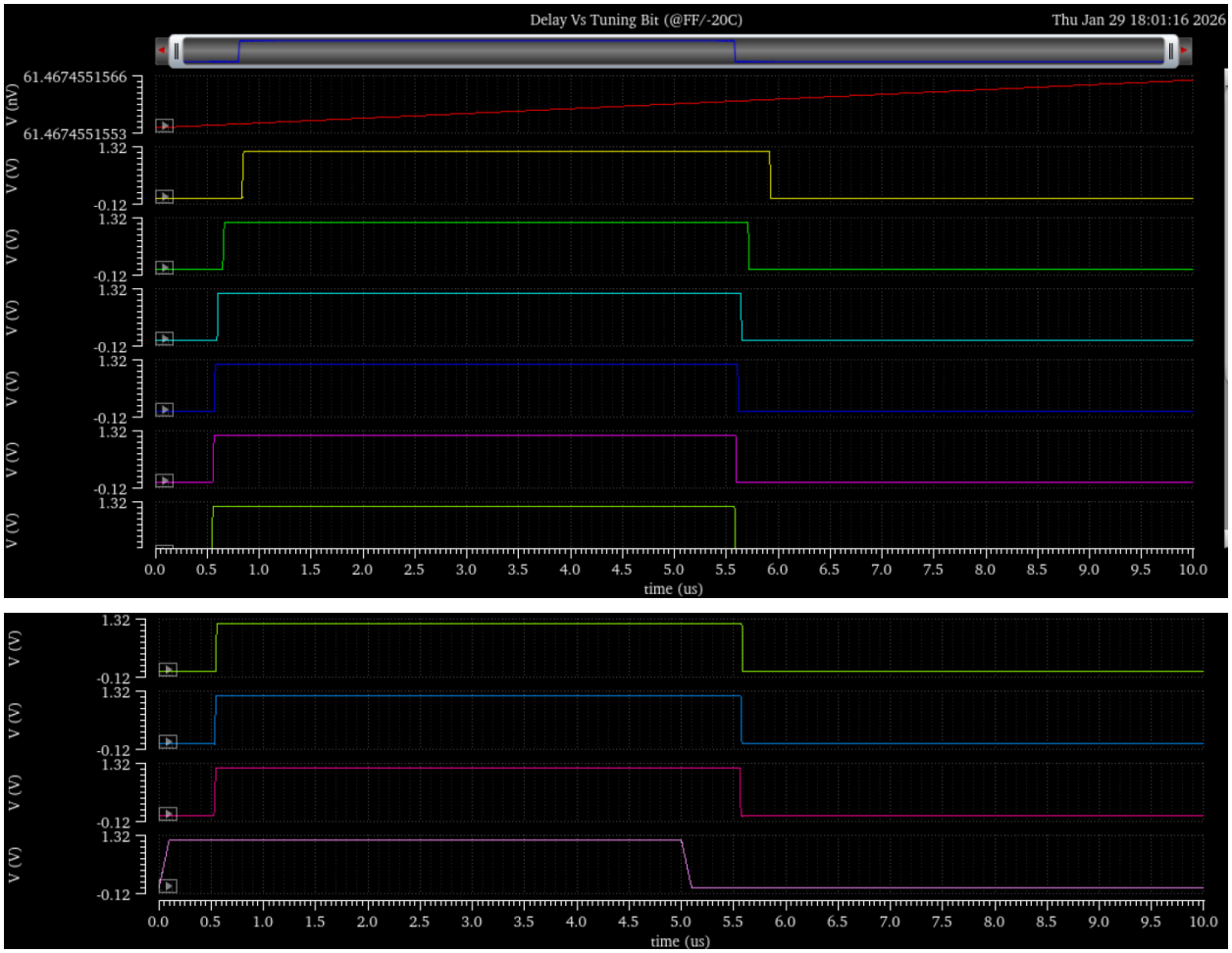


Figure 10: Output of Delay Stages vs Input with varying tuning bits

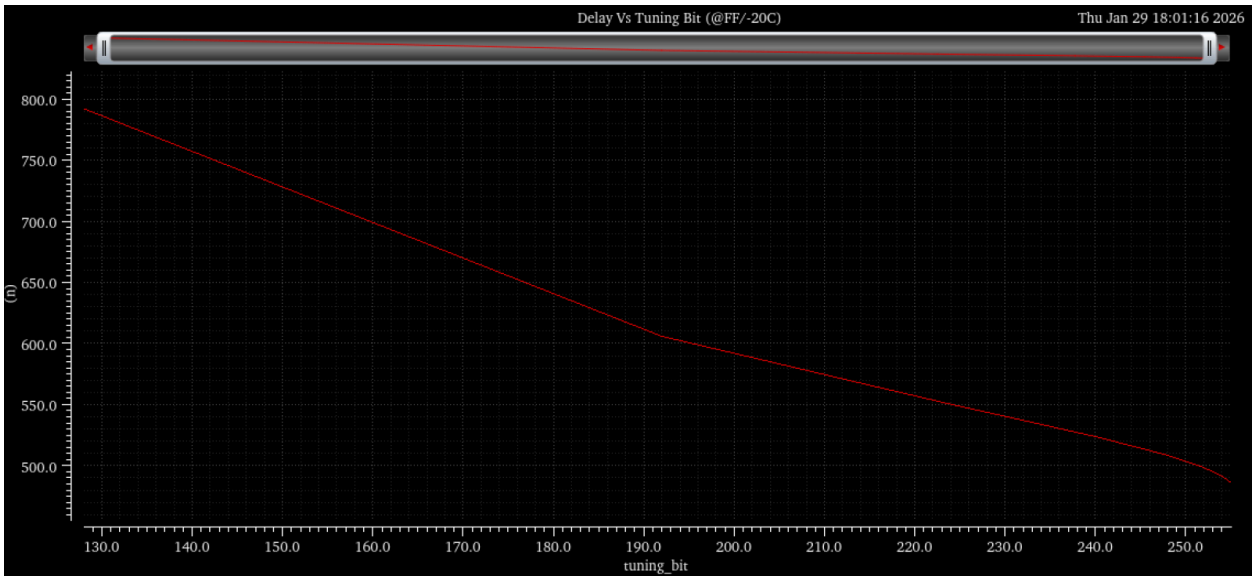


Figure 11: Delay vs Tuning Code

SS 60C

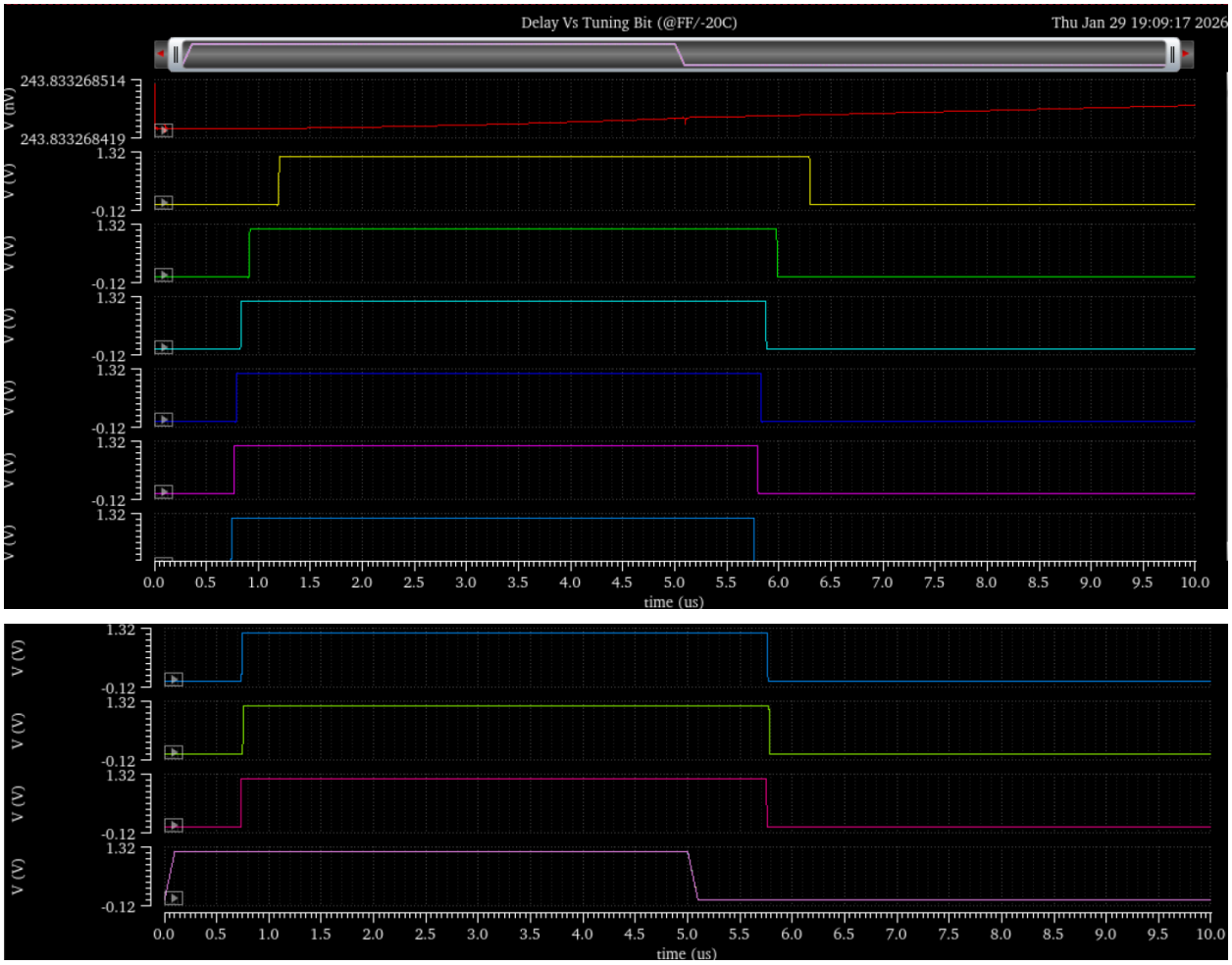


Figure 12: Output of Delay Stages vs Input with varying tuning bits

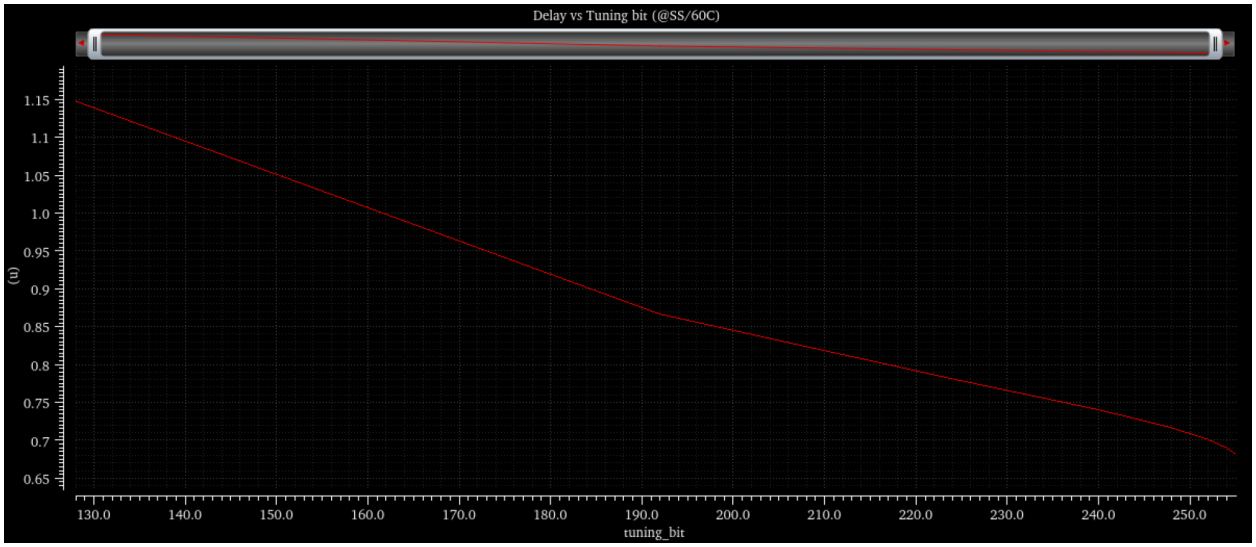


Figure 13: Figure ?: Delay vs Tuning Code

Digital Charge Pump Simulations

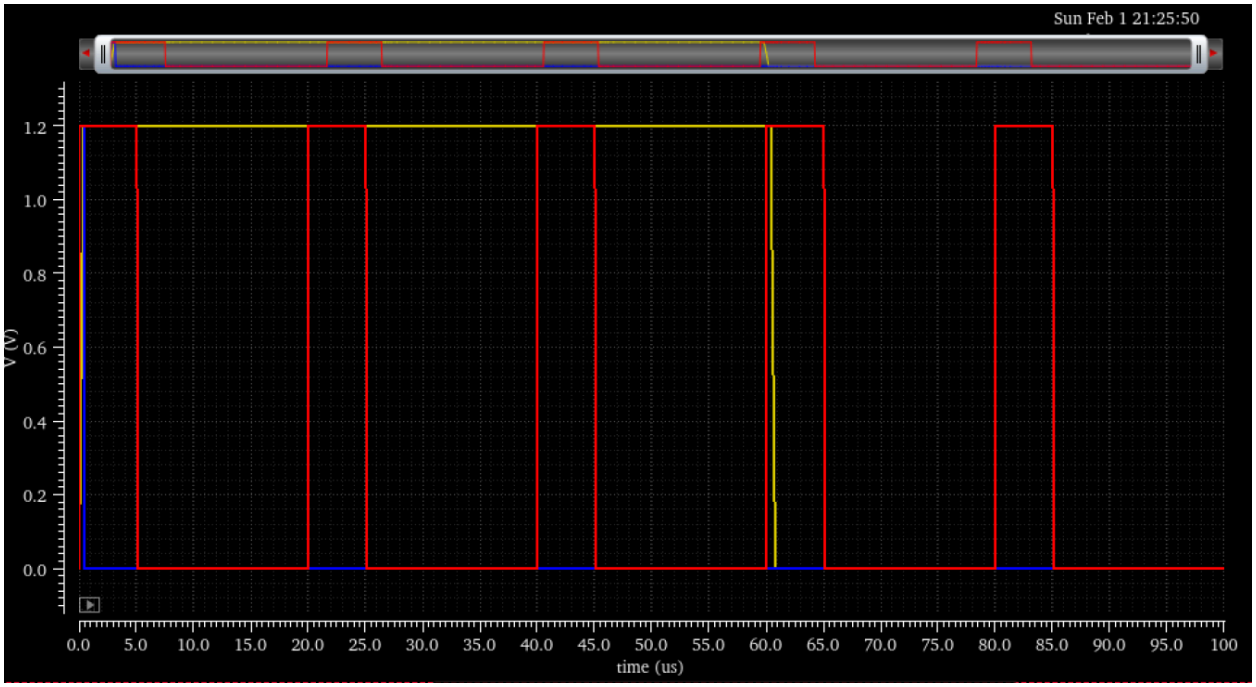


Figure 14: Test Inputs with Red = Triggering, Yellow = Directional, Blue = Reset

@TT/20C

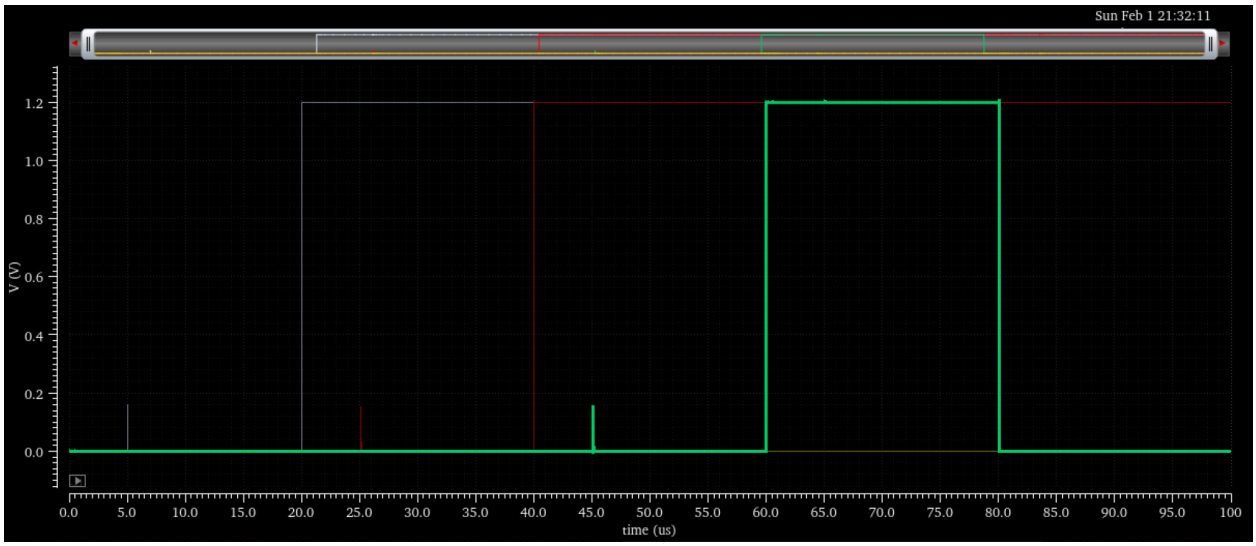


Figure 15: Downwards Counting

@FF/-20C

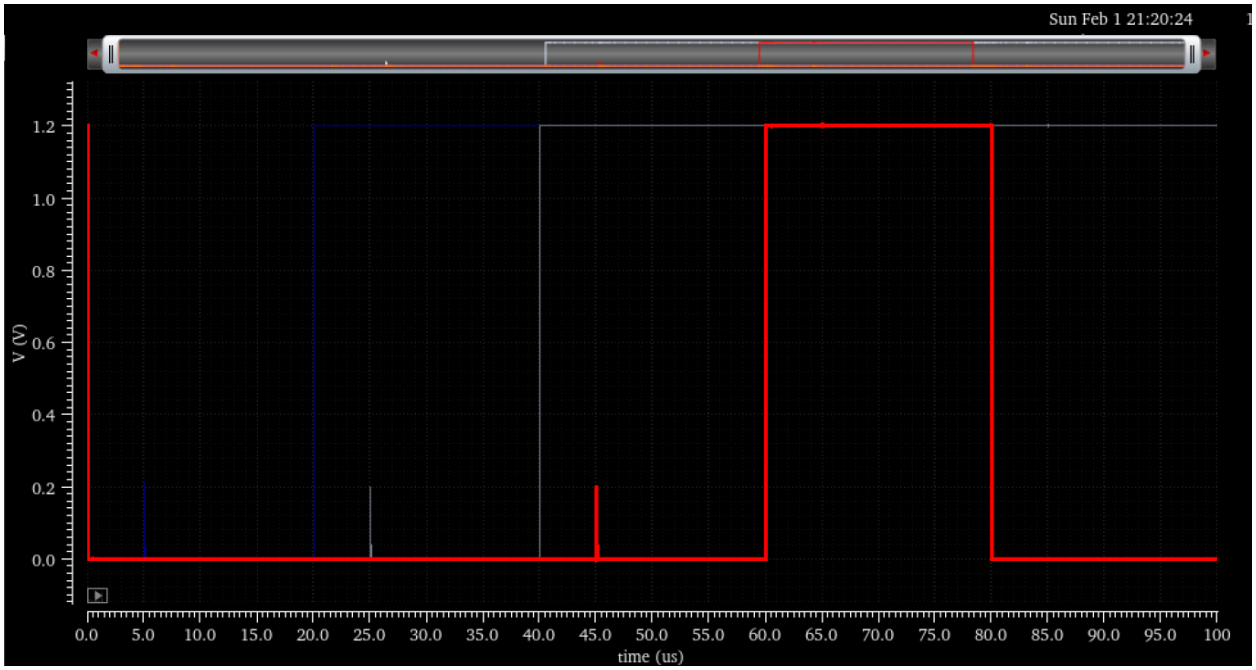


Figure 16: Downwards Counting

@SS/60C

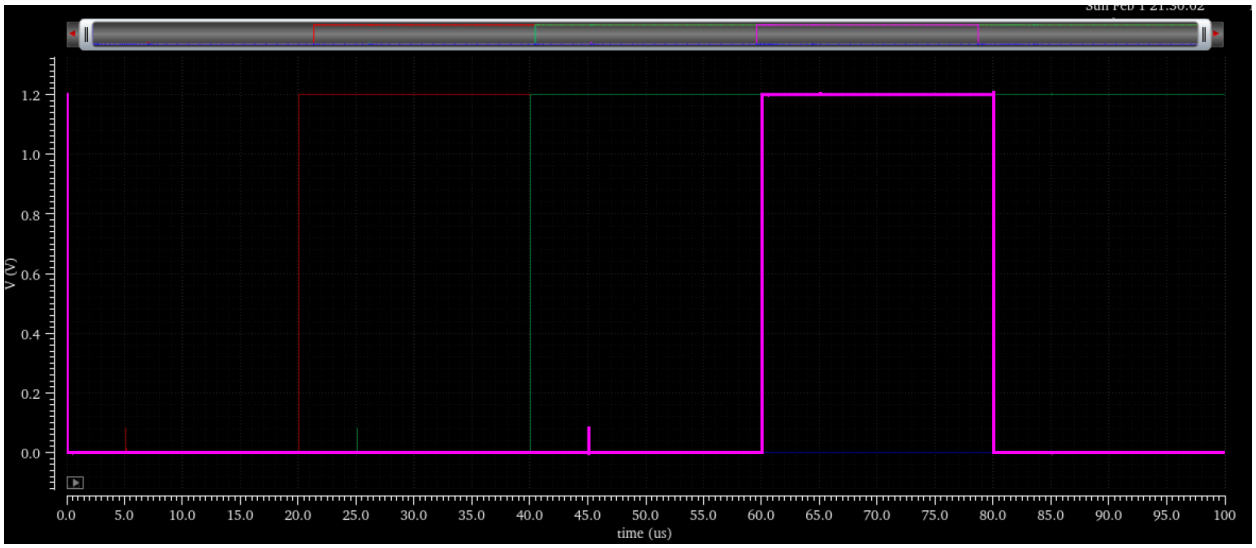


Figure 17: Downwards Counting

DLL Simulations

i)

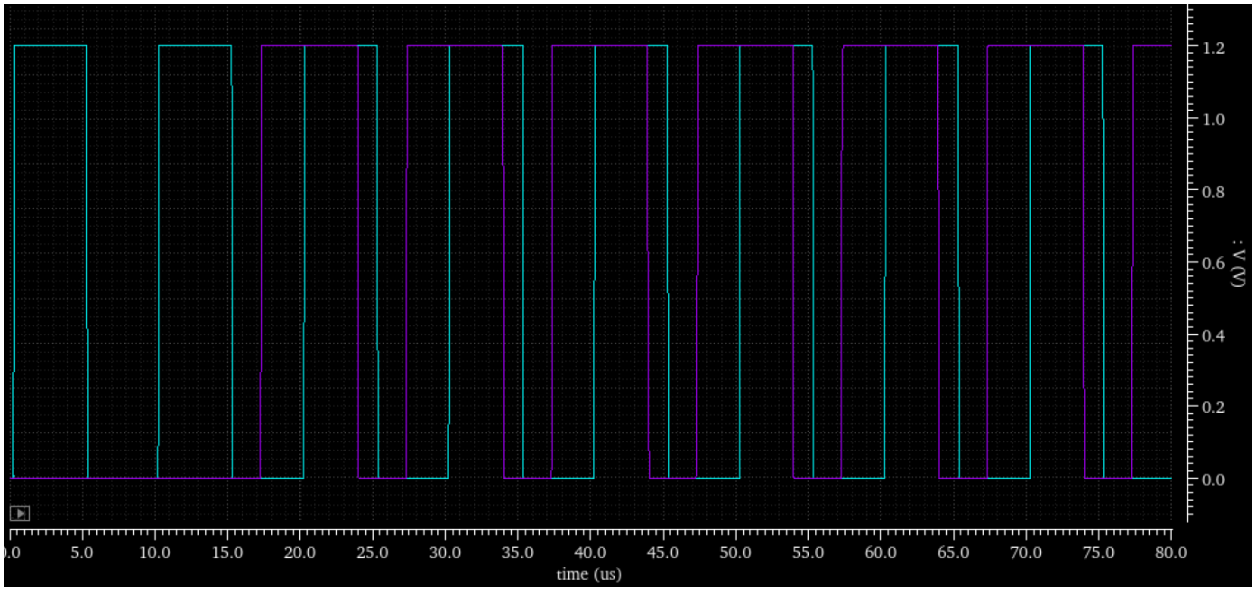


Figure 18: Ref CLK vs DLL Out

ii)

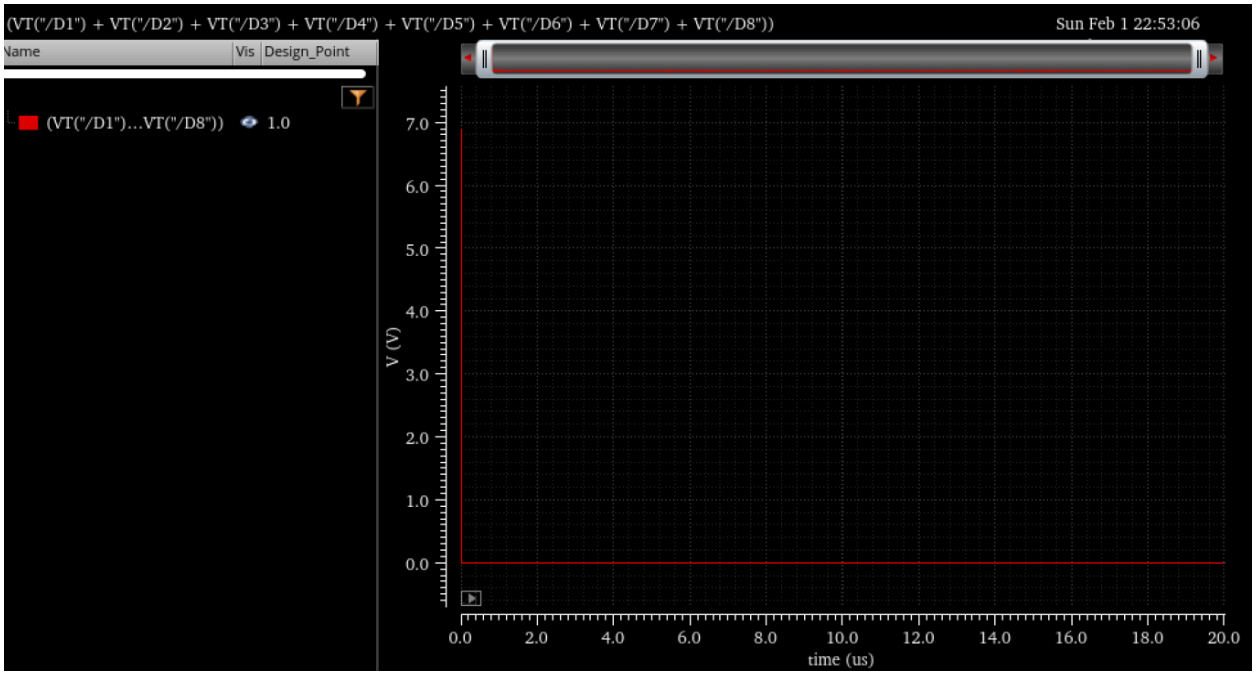


Figure 19: Tuning Bit (Volt) vs Time

AVG Power

ELE827:DLL:1	average(clip((IT(...	-569.3n

Figure 20: Avg Power Consummed

Conclusion

In this lab, a high-frequency clock generator was designed using an All-Digital Delay-Locked Loop (ADDLL). The system utilizes a delay line consisting of 16 delay stages to produce a set of pulses with each stage having a delay of around .625 ns at standard tuning. These delays are then tuned by using a phase detector and up down counter to effectively create a feedback system allowing the tuning bits to be changed depending on the difference between the delay line clk when compared to the ref clk. In this case, the delay can be tuned from ~550 ns to 900 ns at TT/27C with the range varying at a lower operating point at SS/60C and higher at FF/-20CC. The phase detector in this case did not work as intended as the D-Flipflop was not latching on to the signal. This resulted in the feedback loop of the DLL to be broken, never allowing the system to lock as it cannot signal to the up-down counter to change the tuning bits which can be seen in **Figure 19**.